

OpenTelemetry Collector Architecture

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Understanding the OTel Collector Pipeline

The OpenTelemetry Collector is the backbone of OTel deployments—a vendor-agnostic service for receiving, processing, and exporting telemetry data. This notebook covers its architecture, components, and configuration.

Table of Contents

1. Collector Overview
 2. Pipeline Architecture
 3. Receivers
 4. Processors
 5. Exporters
 6. Extensions
 7. Connectors
 8. Configuration Basics
-

Prerequisites

Requirement	Details
Knowledge	OTEL-01: OpenTelemetry Fundamentals
Tools	YAML familiarity for configuration

1. Collector Overview

What is the Collector?

The OpenTelemetry Collector is a standalone service that:

Function	Description
Receives	Ingests telemetry from multiple sources
Processes	Transforms, filters, enriches data
Exports	Sends to one or more backends

Collector Distributions

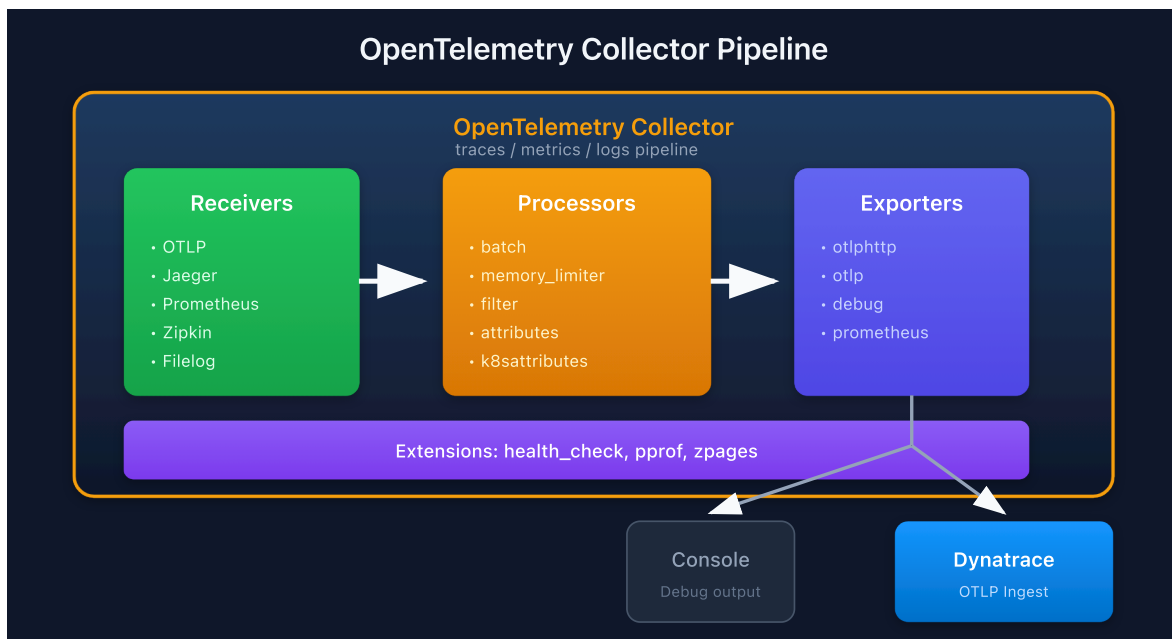
Distribution	Use Case	Components
Core	Minimal, essential only	Basic receivers/exporters
Contrib	Extended functionality	Many community components
Custom	Your specific needs	Build with ocb (OTel Collector Builder)
Vendor	Pre-configured for vendor	Dynatrace, AWS, etc.

Why Use a Collector?

Benefit	Description
Decoupling	Apps don't need backend-specific SDKs
Processing	Centralized filtering, sampling, enrichment
Fan-out	Send to multiple backends
Buffering	Retry and batching for reliability
Security	Centralize credentials and auth

2. Pipeline Architecture

Data Flow



Pipeline Types

Pipeline	Data Type	Example
traces	Distributed traces	Request spans
metrics	Quantitative measurements	CPU usage
logs	Log records	Application logs

Each signal type has its own pipeline(s) with separate receivers, processors, and exporters.

3. Receivers

Receivers ingest telemetry data from external sources.

Common Receivers

Receiver	Protocol	Signals	Use Case
otlp	OTLP (gRPC/HTTP)	Traces, Metrics, Logs	OTel SDKs
jaeger	Jaeger (gRPC/HTTP)	Traces	Jaeger clients
zipkin	Zipkin HTTP	Traces	Zipkin clients
prometheus	Prometheus scrape	Metrics	Prometheus targets
filelog	File tailing	Logs	Log files
hostmetrics	Host stats	Metrics	CPU, memory, disk

OTLP Receiver Configuration

```
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
      http:
        endpoint: 0.0.0.0:4318
```

Prometheus Receiver

```
receivers:
  prometheus:
    config:
      scrape_configs:
        - job_name: 'my-app'
          scrape_interval: 15s
          static_configs:
            - targets: ['app:8080']
```

4. Processors

Processors transform, filter, or enrich telemetry data.

Essential Processors

Processor	Purpose	When to Use
batch	Batch data for efficiency	Always
memory_limiter	Prevent OOM	Always
resourcedetection	Add resource attributes	K8s, cloud
attributes	Add/modify attributes	Enrichment
filter	Drop unwanted data	Cost reduction
transform	OTTL-based transformations	Complex processing

Batch Processor (Critical)

```
processors:
  batch:
    timeout: 10s
    send_batch_size: 1000
    send_batch_max_size: 1500
```

Memory Limiter (Critical)

```
processors:
  memory_limiter:
    check_interval: 1s
    limit_mib: 400
    spike_limit_mib: 100
```

Attributes Processor

```
processors:
  attributes:
    actions:
      - key: environment
        value: production
        action: upsert
      - key: api.key
        action: delete
```

Filter Processor

```
processors:
  filter:
    traces:
      span:
        - 'attributes["http.url"] == "/health"'
```

5. Exporters

Exporters send processed telemetry to backends.

Common Exporters

Exporter	Target	Signals	Notes
otlphttp	OTLP over HTTP	All	Dynatrace, many backends
otlp	OTLP over gRPC	All	Lower overhead
debug	Console/logs	All	Development
prometheus	Prometheus endpoint	Metrics	Pull model
file	Local files	All	Debugging

OTLP HTTP Exporter (Dynatrace)

```
exporters:
  otlphttp:
    endpoint: https://{your-env}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}
```

Debug Exporter

```
exporters:
  debug:
    verbosity: detailed
    sampling_initial: 5
    sampling_thereafter: 200
```

Multiple Exporters (Fan-Out)

```
exporters:
  otlphttp/dynatrace:
    endpoint: https://${DT_ENV_ID}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}
  otlphttp/backup:
    endpoint: https://backup-backend.example.com

service:
  pipelines:
    traces:
      receivers: [otlp]
      processors: [batch]
      exporters: [otlphttp/dynatrace, otlphttp/backup] # Both!
```

6. Extensions

Extensions provide operational capabilities for the Collector itself.

Common Extensions

Extension	Purpose
health_check	Liveness/readiness endpoints
pprof	Go profiling data
zpages	Debug pages for pipelines
memory_ballast	Pre-allocate memory (deprecated)

Configuration

```
extensions:
  health_check:
    endpoint: 0.0.0.0:13133
  pprof:
    endpoint: 0.0.0.0:1777
  zpages:
    endpoint: 0.0.0.0:55679

service:
  extensions: [health_check, pprof, zpages]
```

7. Connectors

Connectors link pipelines, acting as both exporter and receiver.

Use Cases

Connector	Function
<code>spanmetrics</code>	Generate metrics from spans
<code>servicegraph</code>	Build service dependency graph
<code>count</code>	Count data points

Span Metrics Connector

```
connectors:
  spanmetrics:
    dimensions:
      - name: http.method
      - name: http.status_code
    histogram:
      explicit:
        buckets: [100ms, 500ms, 1s, 5s]

service:
  pipelines:
    traces:
      receivers: [otlp]
      exporters: [spanmetrics, otlphttp]
    metrics:
      receivers: [spanmetrics] # Receives from connector
      exporters: [otlphttp]
```

8. Configuration Basics

Minimal Configuration

```
receivers:
  otlp:
    protocols:
      grpc:
        endpoint: 0.0.0.0:4317
      http:
        endpoint: 0.0.0.0:4318

processors:
  batch:
    timeout: 10s

exporters:
  otlphttp:
    endpoint: https://${DT_ENV_ID}.live.dynatrace.com/api/v2/otlp
    headers:
      Authorization: Api-Token ${DT_API_TOKEN}

service:
  pipelines:
    traces:
      receivers: [otlp]
      processors: [batch]
      exporters: [otlphttp]
    metrics:
      receivers: [otlp]
      processors: [batch]
      exporters: [otlphttp]
    logs:
      receivers: [otlp]
      processors: [batch]
      exporters: [otlphttp]
```

Configuration Best Practices

Practice	Reason
Always use <code>batch</code> processor	Efficiency
Always use <code>memory_limiter</code>	Stability
Put <code>memory_limiter</code> first	Fail fast
Use environment variables for secrets	Security
Validate config before deploy	Catch errors early

Summary

In this notebook, you learned:

- Collector overview and distributions
- Pipeline architecture: receivers → processors → exporters
- Common receivers (OTLP, Prometheus, Jaeger)
- Essential processors (batch, memory_limiter, filter)
- Exporters for Dynatrace and other backends
- Extensions for operational capabilities
- Connectors for pipeline linking
- Configuration fundamentals

Next Steps

Next Notebook	Topic
OTEL-03: Collector Deployment	Deployment patterns
OTEL-04: Trace Instrumentation	Instrumenting code

References

- [OTel Collector Documentation](#)
 - [Collector Configuration](#)
 - [Collector Releases](#)
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